

Converse, Inverse & Contrapositive :-

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$	$\neg Q \rightarrow \neg P$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

converse of $P \rightarrow Q$

inverse of $P \rightarrow Q$

contrapositive of $P \rightarrow Q$

$P \rightarrow Q$
 $Q \rightarrow P \rightarrow$ Con
 $\neg P \rightarrow \neg Q \rightarrow$ Inverse
 $\neg Q \rightarrow \neg P \rightarrow$ Contra.

- ↳ A conditional statement and its contrapositive are equivalent.
- ↳ Converse & inverse of a conditional statement are equivalent.
- ↳ Converse or inverse is not equivalent to the implication.

eg. P : It is raining.

Q : The home team wins.

$P \rightarrow Q$: If it is raining, then the home team wins.

Converse. $Q \rightarrow P$: ? If the home team wins, then it is raining.

Inverse $\neg P \rightarrow \neg Q$: ? If it is not raining, then the home team does not win.

Contrapositive $\neg Q \rightarrow \neg P$: ? If the home team does not win, then it is not raining.

6.) Biconditionals :- (iff) ($\leftrightarrow$)

↳ Let p & q be propositions, the biconditional statement, " $p \leftrightarrow q$ ", true when both p & q have the same values.

↳ Different ways expressing biconditionals :-

1. " p iff q "
2. " p if and only if q ".
3. " p is necessary and sufficient for q " → "taking a flight is necessary & sufficient for buying a ticket".
4. "if p then q , and conversely".
5. " p exactly when q ".

eg.
 p : You can take a flight.
 q : You buy a ticket.

$p \leftrightarrow q$: ? You can take a flight if & only if You buy a ticket.

↳ Truth table :-

	p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
→	T	T	T	T	T	T
	T	F	F	F	T	F
	F	T	F	T	F	F
→	F	F	T	T	T	T

↪ equivalent ↪

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv \neg (p \oplus q)$$

1. Tautology :- A proposition which is always true irrespective of the truth values of the underlying variables.

eg. $(P \vee \neg P)$

P	$\neg P$	$(P \vee \neg P)$
T	F	T
F	T	T

→ this is a tautology.

2. Contradiction :- A proposition which is always false irrespective of the truth values of the underlying variables.

eg. $(P \wedge \neg P)$

P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

3. Contingency :- A proposition which is neither a tautology nor a contradiction.

eg. $P \vee Q, P \wedge Q$.

Q. Given the proposition, ① $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$, is it a:

② $(p \rightarrow q) \rightarrow (p \wedge q)$

- A. Tautology.
- B. Contradiction.
- C. Contingency.

Sol.

①.

p	q	$(p \rightarrow q)$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$	$x \leftrightarrow y$	$p \wedge q$	$x \rightarrow z$
T	T	(T)	F	F	(T)	T	T	T
T	F	(F)	T	F	(F)	T	F	T
F	T	(T)	F	T	(T)	T	F	F
F	F	(T)	T	T	(T)	T	F	F

$x \leftrightarrow y$
Is a tautology.

neither tautology
nor a contradiction
 $\therefore$ Contingency.

Precedence of logical operations :-

1. $\neg$
2. $\wedge$
3. $\vee$
4. $\rightarrow$
5. $\leftrightarrow$

↑
High.

eg. $\boxed{P \Rightarrow q \vee r} \equiv P \rightarrow (q \vee r) \checkmark$

$(P \rightarrow q) \vee r \times$

Logical Equivalence :-

↳ X and Y are logically equivalent ($X \equiv Y$), if they have same truth values.

If X is true, then Y is true.

If X is false, then Y is false.

↳ $X \equiv Y$; $X \leftrightarrow Y$ is a tautology $\Leftarrow$

	X P	Y Q	$P \leftrightarrow Q$
①	T	T	T ✓
②	T	F	F
③	F	T	F
④	F	F	T ✓

↳ Standard logically equivalent statements :-

↳ These are useful for simplification while proving the equivalence of large statements.

1. Identity laws :-

$$P \wedge T \equiv P$$

$$P \vee F \equiv P$$

2. Domination laws :

$$P \vee T \equiv T$$

$$P \wedge F \equiv F$$

3. Idempotent laws :- $P \vee P \equiv P$
 $P \wedge P \equiv P$

4. Double negation law :- $\neg(\neg P) \equiv P$

5. Commutative laws :- $P \vee Q \equiv Q \vee P$
 $P \wedge Q \equiv Q \wedge P$

6. Associative laws :- $(P \vee Q) \vee R \equiv P \vee (Q \vee R)$
 $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$

7. Distributive laws :- $\underbrace{P \vee (Q \wedge R)}_{\text{arrow}} \equiv \underbrace{(P \vee Q)}_{\text{circle}} \wedge \underbrace{(P \vee R)}_{\text{circle}}$
 $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

8. Absorption laws :- $\boxed{P \vee (P \wedge Q)} \equiv P$ $\rightarrow P \vee (P \wedge Q)$
 $P \wedge (P \vee Q) \equiv P$ $\Rightarrow (P \vee P) \wedge (P \vee Q)$
 $\Rightarrow P \wedge (P \vee Q)$

9. Negation laws :- $P \vee \neg P \equiv T$
 $P \wedge \neg P \equiv F$

10. De Morgan's law :- I law : $\neg (p \wedge q) \equiv \neg p \vee \neg q$ *

II law : $\neg (p \vee q) \equiv \neg p \wedge \neg q$

eg. p : It is raining.

q : It is windy.

$p \wedge q$: ? It is raining and it is windy.

$\neg (p \wedge q)$: ? It is not true that both it is raining & it is windy.

$\equiv \neg p \vee \neg q$: ? Either it is not raining, or it is not windy.

eg. p : It is snowing.

q : The roads are icy.

$p \vee q$: Either it is snowing or the roads are icy.

$\neg (p \vee q)$: ? It is not true that either it is snowing or the roads are icy.

$\equiv \neg p \wedge \neg q$: ? It is not snowing & the roads are not icy.